

Alcohol Level Tester



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A lot of accidents happen every day due to drunk driving. This is a very useful circuit for testing whether a driver is drunk or not. The circuit is easy to use, inexpensive and indicates various levels of alcohol consumption through LEDs.

Circuit and working

Fig. 1 shows circuit of the alcohol level tester. Alcohol sensor MQ3 is used

here to detect the alcohol fumes' concentration.

Pins 2 and 5 of MQ3 are connected to 5V power supply and the ground, respectively, as shown in the circuit. These pins are actually the heating-coil pins of the sensor. The input pins 1 and 3 of the sensor are also connected to 5V. Pins 4 and 6 are connected to input pin 5 of LM3914 (IC1).

The number of glowing LEDs indicates the concentration level of alcohol

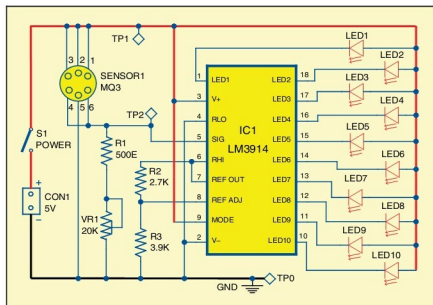


Fig. 1: The alcohol level tester's circuit diagram

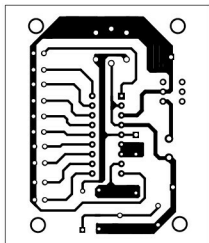


Fig. 2: An actual-size, single-side PCB layout for the circuit

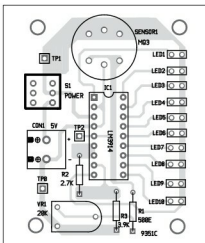


Fig. 3: Component layout for the PCB

PARTS LIST

Semiconductors:

- IC1 - LM3914 bar graph
- LED1-LED10 - Flat LED
- Resistors (all 1/4-watt, $\pm 5\%$ carbon):
- R1 - 500-ohm
- R2 - 2.7-kilo-ohm
- R3 - 3.9-kilo-ohm
- VR1 - 20-kilo-ohm preset

Miscellaneous:

- CON1 - 2-pin terminal connector
- S1 - On/off switch
- SENSOR1 - MQ3 alcohol sensor

Test Points

Test point	Details
TP0	0V
TP1	5V
TP2	MQ3 sensor output

detected by the sensor. Resistor R1 and preset VR1 are used to calibrate the output voltage from the sensor.

When a drunk breathes out near sensor1, the alcohol vapours come in contact with MQ3 sensor and its resistance changes. This raises signal level at pin 5 of IC1 a larger number of LEDs start glowing than the two that were glowing before.

Construction and testing

An actual-size, single-side PCB for the alcohol level tester is shown in Fig. 2 and its component layout in Fig. 3. After assembling the circuit on PCB, enclose it in a suitable case.

Switch on the circuit and verify the test points mentioned in the table. Before using the circuit, vary VR1 until LED1 and LED2 glow when the sensor is in normal environment. Now open the cork of an alcohol bottle and bring it near sensor1. You will notice that LED3, LED4 and LED5 also start glowing. As you bring the alcohol bottle's open mouth very close to sensor1, all the LEDs (LED1 through LED10) will start glowing.

We do not recommend that you actually consume some alcohol to test the circuit. The above test with a bottle of alcohol, or its sample in a small glass, should suffice for most purposes. Until you find a drunk, of course! ●

The author is an electronics hobbyist